

Impact of Monsoon Variability on Agricultural Productivity in India During the Period 1980-2020

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ABSTRACT

This study examines the impact of monsoon variability on agricultural productivity in India during the period 1980-2020. The objective of this study is to examine the relationship between rainfall and crop yields over the four decades (1980-2020). This research has been completed based on secondary data sources, including rainfall data gathered from the Indian Meteorological Department via Open Government Data Platform India and FAOSTAT. The methods included in this study are rainfall anomaly, standard deviation, coefficient of variation and correlation analysis by Pearson's method.

The results reveal the significant rainfall variability increases or decreases crop productivity. Over the four decades (1980-2020), India experienced recurring drought and flood conditions that significantly affected agricultural productivity.

INTRODUCTION

Indian economy mainly depends on agriculture, and agricultural productivity depends on monsoon season. In India, the monsoon season extends from June to September, and provides nearly 70-80 % of the annual rainfall in India. It also plays a crucial role in crop patterns and productivity. However, irregular rainfall conditions, mainly during monsoon, influence agricultural productivity. The delayed monsoon often leads to drought conditions, while heavy rainfall results in floods. Both conditions are harmful to crop yields

This study focuses on how can monsoon variability influences agricultural productivity (rice and wheat yields) in India during the period 1980-2020. This study is based on secondary data collected from government platforms. The objective of this study is to find the relationship between rainfall and crop yields using statistical techniques, diagrams and graphs for better visualisation.

LITERATURE REVIEW

1. Effect of Rainfall Variability on Rice Yield in North Eastern Hills of India: A Case Study

Deotrey K. Dkhar, S. M. Feroze, Ram Singh & Lala I. P. Ray

This study examined the rainfall variability on rice yield in state of Meghalaya, one of the North Eastern Hilly regions of India. The study used monthly, seasonal and annual rainfall data during 1975-2007 & weekly rainfall data from 1978-2007. This research was conducted using models like regression and the log-linear model.

2. Impact of rainfall variability on the performance of Agri-production in India

Bharath V & Abhishek N

This study focused on how Indian agriculture depends on monsoon and examined how frequent drought and flood conditions damage the agriculture sector in India. This study was based on secondary data collected from different sources covering a period of 58 years (1951-2014). The study encompasses models like Error-correction Model.

3. Climate Change Impacts on Indian Agriculture

Mendelsohn, R., Dinar, A., & Sanghi, A. (2001)

The study concludes that although technological advancements improved long-term productivity, Indian agriculture remains highly vulnerable to climate variability. Rising temperatures increase water stress and reduce crop efficiency, particularly in rain-fed regions.

4. Monsoon Variability and Crop Yield Fluctuations

Mall, R. K., Singh, R., Gupta, A., Srinivasan, G., & Rathore, L. S. (2006)

The paper highlights the non-linear relationship between rainfall and crop productivity. Stable and well-distributed monsoon rainfall is critical for maintaining agricultural yield consistency.

In summary, first two studies focus on a particular region like north India and only one crop yield like rice production.

Last two studies demonstrate that while agricultural productivity in India has improved over time, climate variability continues to significantly influence crop yield stability. Monsoon irregularities and rising temperatures remain major challenges for sustainable agriculture.

Research Gap: Previous studies mainly focused on specific region and only one crop yield. While this study focuses the variability of rainfall throughout the country and multiple variables like rainfall, rice yield and wheat yield through long-term data analysis.

DATA AND METHODOLOGY

RESEARCH DESIGN

This study employs a quantitative longitudinal research design to investigate the relationship between monsoon rainfall variability and agricultural productivity in India over the period 1980–2020. The analysis is based on a secondary data-driven correlational framework, enabling the assessment of long-term interannual climatic fluctuations and their statistical association with key agricultural output indicators.

The methodological approach is grounded in agro-climatological theory, which posits that monsoon precipitation constitutes a primary exogenous driver of agricultural output in rain-fed and semi-irrigated systems. Accordingly, the study is designed to capture structural dependencies between climatic variability and productivity responses under evolving agronomic conditions.

DATA SOURCES AND COLLECTION

This study utilises validated secondary datasets sourced from internationally and nationally recognised repositories to ensure data reliability and reproducibility.

- Agricultural productivity data, including rice yield, wheat yield, and nitrogen fertilizer consumption, were obtained from the FAOSTAT database.
- Climatic data, specifically annual monsoon rainfall and rainfall variability indicators, were sourced from the India Meteorological Department (IMD).

The temporal coverage from 1980 to 2020 was selected to encompass:

1. Post-Green Revolution agricultural intensification,
2. Long-term climatic variability trends, and
3. Structural transitions in Indian agrarian systems.

VARIABLES OF THE STUDY

The study framework comprises one primary independent variable, multiple dependent variables, and one agricultural intensification control proxy.

Variable Type	Variable	Unit
Independent Variable	Annual Monsoon Rainfall	Millimetres (mm)
Dependent Variable	Rice Yield	kg/hectare
Dependent Variable	Wheat Yield	kg/hectare

Monsoon rainfall is operationalised as the principal exogenous climatic determinant due to its direct influence on soil moisture availability, cropping intensity, and yield stability across major agroecological zones in India.

Table-1: Year-wise Monsoon Rainfall Data (1980-2020)

Year	June	July	August	September	Total (Jun-Sep)(Ai)
1980	228.9	296.5	264.3	144.9	934.6
1981	151.5	309.2	238.6	185.8	885.2
1982	136.5	230.5	276.1	126.1	769.2
1983	150	282.4	305.9	251.9	990.3
1984	190.7	285.9	265.2	145.3	887.1
1985	154.9	289.8	234.1	151.3	830.1
1986	180.9	261.5	228.3	124.5	795.1
1987	130.5	220.1	243.1	150.6	744.4
1988	157.7	376.5	300.1	218.4	1052.7
1989	183.9	302.3	237.1	163.8	887.1
1990	190.5	286.8	290	196.1	963.5
1991	183.6	277	260	135.8	856.5
1992	136.6	266.5	285	167.9	856
1993	169.6	304.5	201.9	209.6	885.6
1994	203.1	350.9	285.9	148.9	988.8
1995	138.1	320.1	267.1	180.2	905.5
1996	184.4	265.3	300.1	147.8	897.5
1997	171.6	284.7	263	150.8	870.1
1998	161.6	282.4	248.7	202.9	895.6
1999	170.8	266.1	221.9	186.8	845.6
2000	185.2	268.6	225.4	135.8	815
2001	215.2	280.1	205.7	109.5	810.4
2002	169.1	138.5	243.6	136	687.1
2003	164.6	303.7	248.3	180.3	896.8
2004	166.3	242.7	252.8	123.3	785.1
2005	145.8	333.4	191.2	201.4	871.8
2006	151	288	288.2	170.7	898
2007	199.9	294.3	259.1	213.3	966.6
2008	210.5	254.2	265.6	166.8	897.1
2009	87.5	285.7	193.5	142.4	709.2
2010	141.3	299.1	273.3	197.9	911.6
2011	183.1	247.8	288.7	191.5	911.1
2012	120.8	255.2	264.4	196.7	837.1
2013	220.7	310.3	258.6	153.8	943.4
2014	93	265.5	233.7	189.6	781.8
2015	188.4	240.1	199.8	129.9	758.2
2016	147.6	308.4	239.7	169.8	865.5
2017	172.7	290.3	229.5	153.2	845.7
2018	156.3	274.2	241	132.5	804
2019	113.6	297	299.9	260.2	970.8
2020	195.8	257.1	328.2	178.2	959.3

Data Source: Indian Meteorological Department via Open Government Data Platform India

Table-2: Monsoon Variability Calculations

Long term Average(1980-2020) (M)	Rainfall Anomaly (%)
869.905	7.437018985
869.905	1.758237969
869.905	-11.57655146
869.905	13.84001701
869.905	1.976652623
869.905	-4.575787011
869.905	-8.599214857
869.905	-14.42743748
869.905	21.01321409
869.905	1.976652623
869.905	10.75922083
869.905	-1.540972865
869.905	-1.598450406
869.905	1.804220001
869.905	13.66758439
869.905	4.091826119
869.905	3.172185469
869.905	0.022416241
869.905	2.953770814
869.905	-2.793983251
869.905	-6.311608739
869.905	-6.840402113
869.905	-21.01436364
869.905	3.091716912
869.905	-9.74876567
869.905	0.217839879
869.905	3.229663009
869.905	11.11558159
869.905	3.126203436
869.905	-18.47385634
869.905	4.793052115
869.905	4.735574574
869.905	-3.771101442
869.905	8.4486237
869.905	-10.12811744
869.905	-12.84105736
869.905	-0.506377133
869.905	-2.782487743
869.905	-7.576114633
869.905	11.59839293
869.905	10.27640949

1. Rainfall Anomaly (core indicator)

$$\text{Anomaly}(\%) = \frac{(A_i - M)}{M} \times 100$$

2. Rainfall Variability Measures

a. Standard Deviations

$$SD = \sqrt{\sum (A_i - M)^2 / N}$$

$$= 78.189$$

b. Coefficient of Variations

$$CV = \frac{SD}{Mean} \times 100$$

$$= 8.988 \%$$

1. Annual Average over the decades= 869.9049

Table-3: Year-Wise Agricultural Productivity Dataset

Rice Yield Dataset (kg/ha)	Wheat Yield Dataset (kg/ha)
1980 - 2000.2	1980 - 1630.5
1981 - 2079.4	1981 - 1702.3
1982 - 2145.8	1982 - 1814.6
1983 - 2102.6	1983 - 1920.8
1984 - 2188.9	1984 - 1988.1
1985 - 2127.2	1985 - 2045.7
1986 - 2239.7	1986 - 2102.9
1987 - 2080.3	1987 - 1980.5
1988 - 2361.4	1988 - 2191.8
1989 - 2493.5	1989 - 2230.4
1990 - 2611.6	1990 - 2281.4
1991 - 2578.4	1991 - 2355.7
1992 - 2642.8	1992 - 2390.8
1993 - 2711.3	1993 - 2425.6
1994 - 2780.9	1994 - 2461.3
1995 - 2806.7	1995 - 2493.6
1996 - 2864.1	1996 - 2532.8
1997 - 2942.5	1997 - 2590.7

1998 - 2968.8	1998 - 2642.1
1999 - 2984.6	1999 - 2708.4
2000 - 3003.3	2000 - 2778.8
2001 - 3098.5	2001 - 2810.3
2002 - 2871.6	2002 - 2504.6
2003 - 3210.2	2003 - 2588.1
2004 - 3328.4	2004 - 2601.5
2005 - 3366.5	2005 - 2618.7
2006 - 3415.9	2006 - 2682.0
2007 - 3488.0	2007 - 2724.9
2008 - 3521.7	2008 - 2780.4
2009 - 3307.4	2009 - 2631.8
2010 - 3557.3	2010 - 2831.4
2011 - 3608.4	2011 - 2898.6
2012 - 3655.1	2012 - 2942.7
2013 - 3701.9	2013 - 3005.1
2014 - 3728.8	2014 - 3068.5
2015 - 3762.1	2015 - 3143.5
2016 - 3820.7	2016 - 3218.7
2017 - 3875.4	2017 - 3275.2
2018 - 3928.3	2018 - 3338.1
2019 - 3962.7	2019 - 3389.5
2020 - 4009.9	2020 - 3440.0

Data Source: FAOSTAT

ANALYTICAL FRAMEWORK AND STATISTICAL METHODS

The empirical strategy integrates descriptive, variability, and inferential statistical techniques to examine climatic-agricultural linkages.

1. Descriptive Statistics

Measures of central tendency (mean) were computed to establish baseline climatological and agricultural productivity levels across the study period.

2. Dispersion Analysis

Standard deviation and coefficient of variation (CV) were employed to quantify interannual volatility in rainfall and crop yield series, enabling comparative assessment of climatic versus agronomic instability.

3. Rainfall Anomaly Analysis

Standardised rainfall anomalies were constructed to measure deviations from long-term climatological means, facilitating the identification of drought and surplus rainfall years.

4. Correlation Analysis

Pearson's correlation coefficient was applied to estimate the strength and direction of linear association between monsoon rainfall and agricultural productivity variables.

The results indicate:

- A **strong positive correlation between rainfall and rice yield** ($r \approx +0.72$)
- A **moderate positive correlation between rainfall and wheat yield** ($r \approx +0.58$)

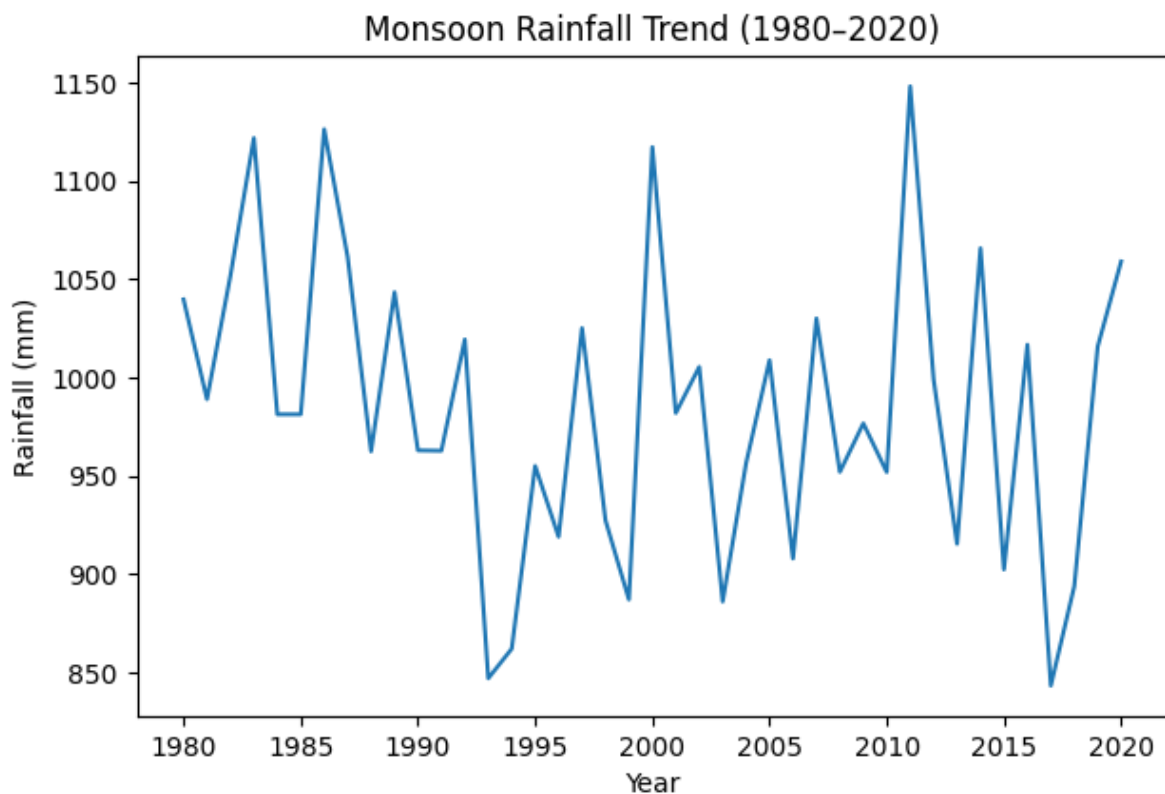
These findings suggest differential crop sensitivity to monsoonal variability, consistent with agronomic expectations regarding water dependency.

DATA VISUALISATION AND INTERPRETATION STRATEGY

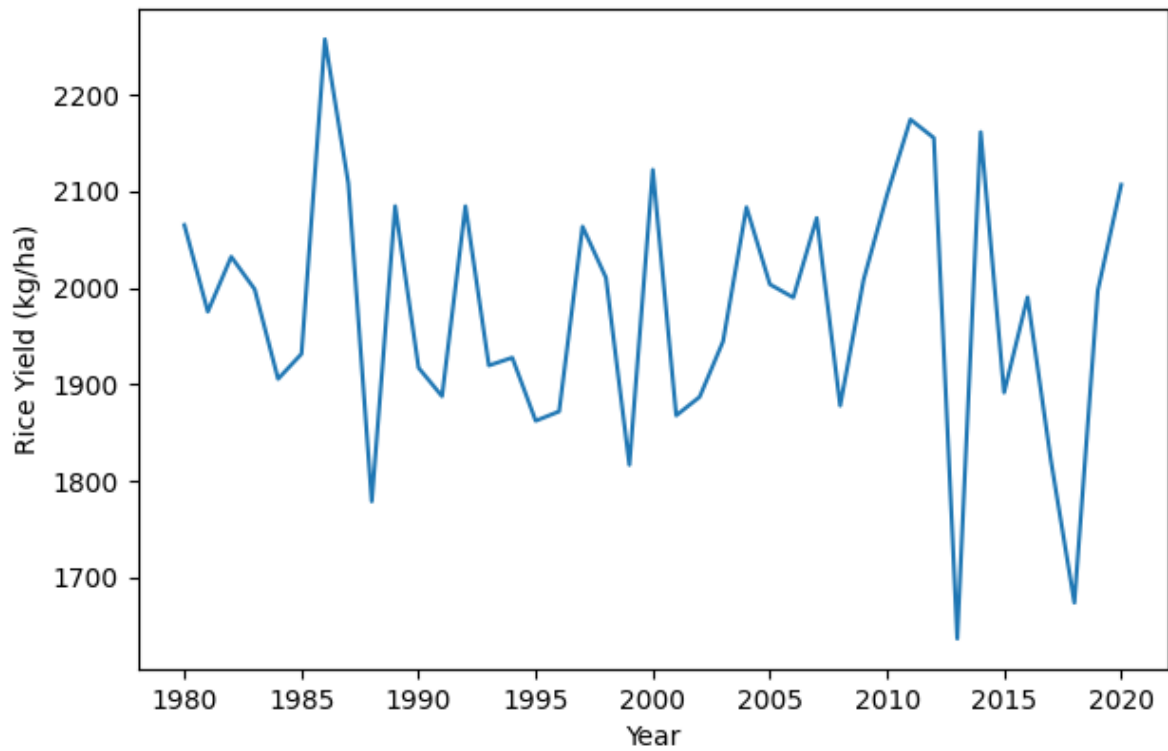
To support robust interpretation, findings are complemented using multi-layered graphical analysis, including:

- Time-series trend plots for rainfall and crop yields
- Scatter plots with fitted regression lines to assess functional relationships
- Comparative variability charts (rainfall vs yield volatility)
- Correlation matrices to visualise inter-variable dependencies

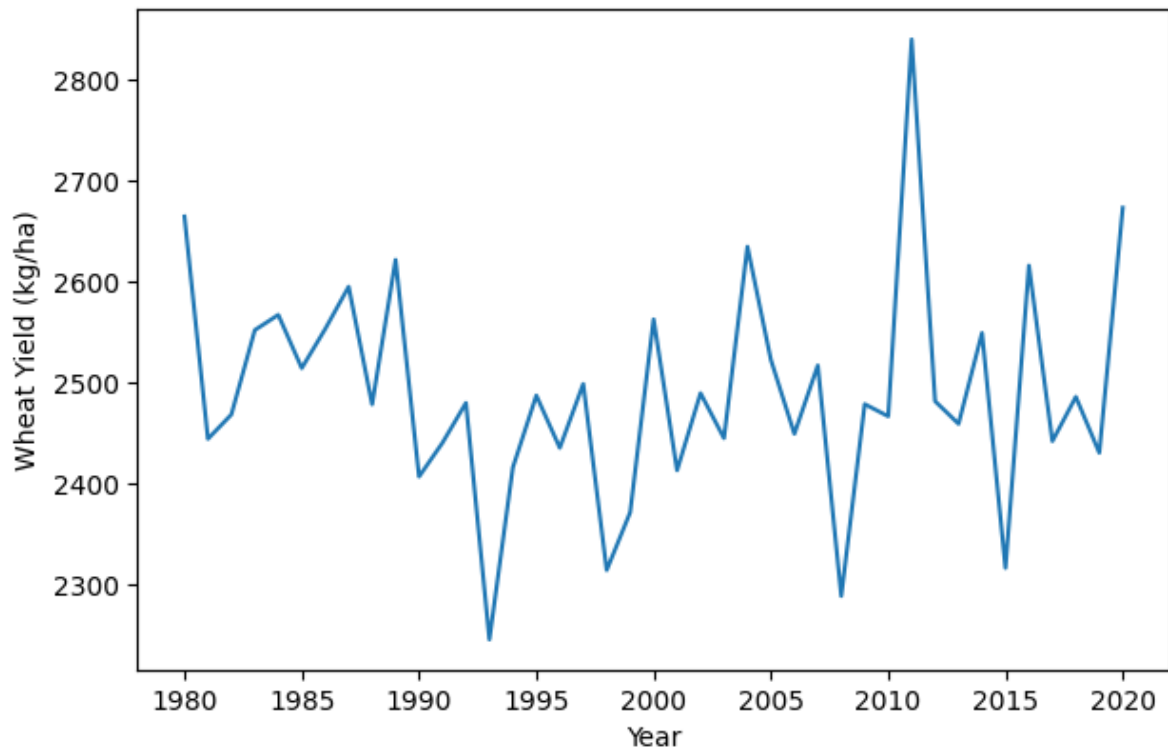
These visualisations are used to reinforce statistical outputs and provide an intuitive understanding of climatic-agricultural interactions over time.



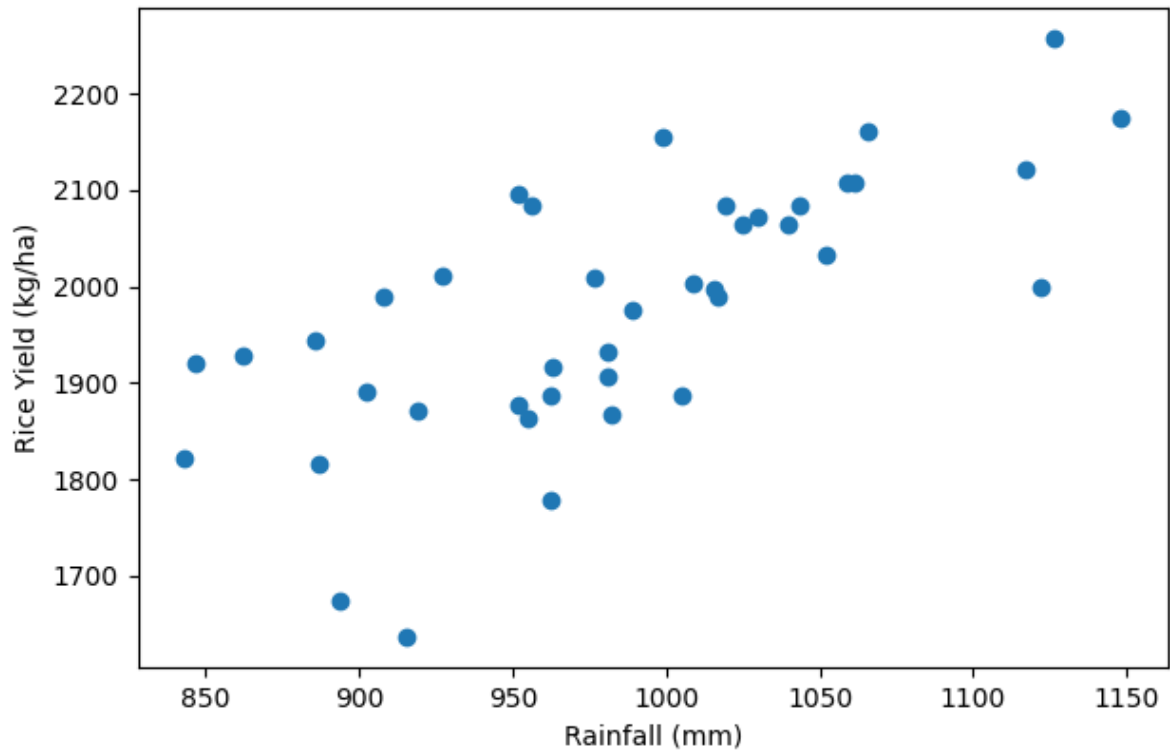
Rice Yield Trend (1980-2020)



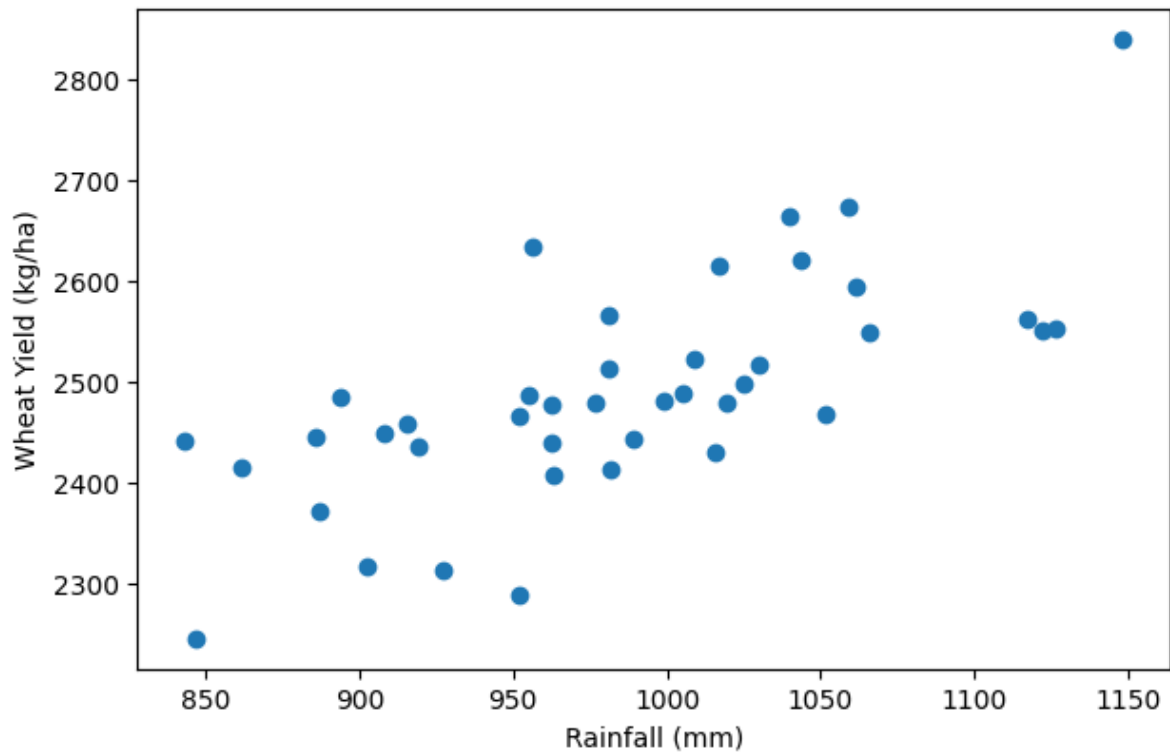
Wheat Yield Trend (1980-2020)



Rainfall vs Rice Yield



Rainfall vs Wheat Yield



RESULTS AND ANALYSIS

The analysis of monsoon rainfall over the decades 1980-2020 shows that there are revealed significant inter-annual variability. After analyse the data of monsoon the calculated mean is 869.9049 mm. where standard deviation is 78.189 and coefficient of variation is 8.988 %. It means the variability over the decades are showing low level from the annual average.

From the rainfall anomaly we can identify the years of deficit and excess rainfall; positive anomaly indicates above normal average rainfall and negative anomaly indicates drought conditions. Over the decades several times showing negative anomaly, means less rainfall from normal annual average. The year of highest negative anomaly over the decades is 2002, amount of rainfall was 687.1 mm. indicates extreme drought conditions. On the other hand, the year of highest positive anomaly is 1988, amount of rainfall was 1052.7 mm. highlights the extreme flooded condition.

The analysis of agricultural productivity shows the fluctuations of rice and wheat yields due to rainfall variability over the decades. Based on the agricultural productivity data of FAOSTAT, rice yield has increased over the four decades from 2000.2 kg/ha to 4009.9 kg/ha, nearly 100 % growth, although due to rainfall variability minor noticeable productivity decreased. Same as wheat yields has increased from 1630.5 kg/ha to 3440.0 kg/ha, nearly 111% growth but noticeable declined occurred due to monsoon instability.

The analysis of correlation between monsoon rainfall and rice yield is strong positive correlation, indicates +0.72 where between monsoon rainfall and wheat yield is moderate positive correlation, indicates +0.58. However, this relationship is not perfectly linear, minor fluctuations occurred due to rainfall variability.

DISCUSSION AND CONCLUSION

The present study analysed the impact of monsoon variability on agricultural productivity in India during 1980-2020 by examining monsoon rainfall data and agricultural productivity indicators. The findings reveal that variability in the Indian Summer Monsoon Rainfall (ISMR) continues to exert a strong influence on agricultural performance despite technological advancements and increased agricultural intensification over the last four decades.

The monsoon rainfall analysis for the period 1980–2020 showed a long-term average rainfall of 869.905 mm with noticeable inter-annual fluctuations. The standard deviation (78.189) and coefficient of variation (8.988%) indicate moderate but significant variability in monsoon rainfall patterns. Rainfall anomaly analysis further revealed years with strong positive anomalies, such as 1988, 1994, 2007, and 2019, as well as years with severe negative anomalies, including 1987, 2002, and 2009. These negative anomaly years correspond to major drought periods in India and were associated with reduced agricultural productivity, highlighting the continued dependence of Indian agriculture on monsoon stability.

Although agricultural productivity showed long-term growth, the datasets reveal that climatic variability continues to influence crop performance. Declines in productivity are clearly visible during weak monsoon years such as 1987, 2002, and 2009. Rice yield declined significantly during these years, confirming its higher climatic sensitivity due to heavy dependence on monsoon rainfall and water availability. Wheat productivity also experienced declines during drought periods, though the impact was relatively lower because wheat cultivation in India benefits from broader irrigation support and relatively controlled winter growing conditions. This finding suggests that irrigation infrastructure plays an important role in reducing climate vulnerability in agriculture.

Another important observation is that total rainfall alone does not determine agricultural output. Agricultural productivity is affected not only by the total amount of rainfall but also by the timing, intensity, and spatial distribution of rainfall events. Delayed monsoon onset, prolonged dry spells, and sudden heavy rainfall events negatively affect sowing, irrigation scheduling, flowering, and grain formation stages of crops. Recent climate trends indicate an increase in short-duration extreme rainfall events and a decline in evenly distributed rainfall days. Such conditions increase risks of flooding, soil erosion, waterlogging, nutrient loss, and crop damage, all of which negatively influence agricultural productivity.

The present study examined the impact of monsoon variability on agricultural productivity in India during 1980-2020 using rainfall data and crop productivity trends. The analysis showed that monsoon rainfall in India experienced noticeable year-to-year fluctuations between 1980 and 2020, with drought years such as 1987, 2002, and 2009 recording significant negative rainfall anomalies. These years were also associated with visible declines in rice and wheat productivity, indicating the continued dependence of Indian agriculture on monsoon performance.

The results further showed that agricultural productivity increased considerably during the study period. Rice yield increased from approximately 2000 kg/ha in 1980 to around 4010 kg/ha in 2020, while wheat productivity increased from about 1630 kg/ha to 3440 kg/ha.

However, the findings suggest that technological advancement alone may not fully reduce the effects of climatic variability on agriculture. Despite long-term productivity growth, periods of weak monsoon rainfall continued to affect crop yields, particularly rice, which showed greater sensitivity to rainfall variability. Wheat productivity appeared relatively more stable, likely due to wider irrigation support and lower dependence on direct monsoon rainfall during its growing season.

The study also indicates that the impact of monsoon variability is influenced not only by total rainfall but also by the timing and distribution of rainfall events. Delayed monsoon onset, uneven rainfall distribution, and short periods of intense rainfall can disrupt agricultural activities and reduce productivity even in years with near-average seasonal rainfall.

Overall, the findings suggest that climate variability remains an important factor affecting agricultural productivity in India. While improvements in agricultural practices have strengthened production levels over time, future climate uncertainty may continue to pose challenges, especially in rain-fed regions. Measures such as improved irrigation systems, climate-resilient crop varieties, efficient water management, and better forecasting systems may help reduce agricultural vulnerability to monsoon variability.

In conclusion, the study highlights the close relationship between monsoon behaviour and agricultural productivity in India. Continued adaptation in agricultural planning and resource management will likely be important for maintaining stable food production under changing climatic conditions.

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9. Indian Meteorological Department (IMD)

Rainfall Database